

BC636, BC636-16, BC638, BC640, BC640-16

High Current Transistors

PNP Silicon



ON Semiconductor

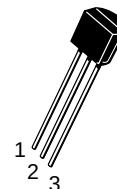
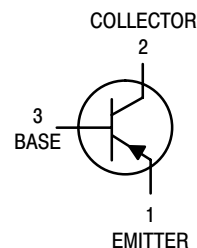
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MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Collector-Emitter Voltage BC636 BC638 BC640	V_{CEO}	-45 -60 -80	Vdc
Collector-Base Voltage BC636 BC638 BC640	V_{CBO}	-45 -60 -80	Vdc
Emitter-Base Voltage	V_{EBO}	-5.0	Vdc
Collector Current — Continuous	I_C	-0.5	Adc
Total Device Dissipation @ $T_A = 25^\circ\text{C}$ Derate above 25°C	P_D	625 5.0	mW mW/ $^\circ\text{C}$
Total Device Dissipation @ $T_C = 25^\circ\text{C}$ Derate above 25°C	P_D	1.5 12	Watts mW/ $^\circ\text{C}$
Operating and Storage Junction Temperature Range	T_J, T_{stg}	-55 to +150	$^\circ\text{C}$

THERMAL CHARACTERISTICS

Characteristic	Symbol	Max	Unit
Thermal Resistance, Junction to Ambient	$R_{\theta JA}$	200	$^\circ\text{C/W}$
Thermal Resistance, Junction to Case	$R_{\theta JC}$	83.3	$^\circ\text{C/W}$



CASE 29
TO-92
STYLE 14

ORDERING INFORMATION

Device	Package	Shipping
BC636	TO-92	5000 Units/Box
BC636ZL1	TO-92	2000/Ammo Pack
BC636-16ZL1	TO-92	2000/Ammo Pack
BC638	TO-92	5000 Units/Box
BC638ZL1	TO-92	2000/Ammo Pack
BC640	TO-92	5000 Units/Box
BC640ZL1	TO-92	2000/Ammo Pack
BC640-16	TO-92	5000 Units/Box

BC636, BC636–16, BC638, BC640, BC640–16

ELECTRICAL CHARACTERISTICS (T_A = 25°C unless otherwise noted)

Characteristic	Symbol	Min	Typ	Max	Unit
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OFF CHARACTERISTICS

Collector–Emitter Breakdown Voltage (I _C = –10 mA, I _B = 0)	BC636 BC638 BC640	V _{(BR)CEO}	–45 –60 –80	— — —	— — —	V _{dc}
Collector–Base Breakdown Voltage (I _C = –100 μA, I _E = 0)	BC636 BC638 BC640	V _{(BR)CBO}	–45 –60 –80	— — —	— — —	V _{dc}
Emitter–Base Breakdown Voltage (I _E = –10 μA, I _C = 0)		V _{(BR)EBO}	–5.0	—	—	V _{dc}
Collector Cutoff Current (V _{CB} = –30 V, I _E = 0) (V _{CB} = –30 V, I _E = 0, T _A = 125°C)		I _{CBO}	— —	— —	–100 –10	nA μA

ON CHARACTERISTICS (1)

DC Current Gain (I _C = –5.0 mA, V _{CE} = –2.0 V) (I _C = –150 mA, V _{CE} = –2.0 V)	BC636 BC636–16 BC638 BC640 BC640–16	h _{FE}	25 40 100 40 40 100 25	— — — — — — —	— 250 250 160 160 250 —	—
Collector–Emitter Saturation Voltage (I _C = –500 mA, I _B = –50 mA)		V _{CE(sat)}	— —	–0.25 –0.5	–0.5 —	V _{dc}
Base–Emitter On Voltage (I _C = –500 mA, V _{CE} = –2.0 V)		V _{BE(on)}	—	—	–1.0	V _{dc}

DYNAMIC CHARACTERISTICS

Current–Gain — Bandwidth Product (I _C = –50 mA, V _{CE} = –2.0 V, f = 100 MHz)		f _T	—	150	—	MHz
Output Capacitance (V _{CB} = –10 V, I _E = 0, f = 1.0 MHz)		C _{ob}	—	9.0	—	pF
Input Capacitance (V _{EB} = –0.5 V, I _C = 0, f = 1.0 MHz)		C _{ib}	—	110	—	pF

1. Pulse Test: Pulse Width ≤ 300 μs, Duty Cycle 2.0%.

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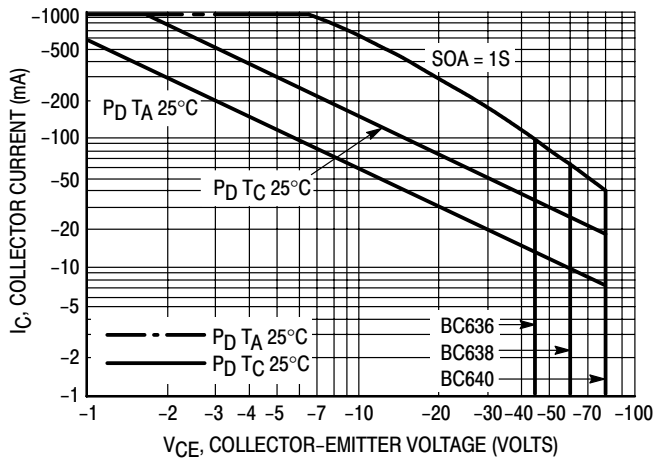


Figure 1. Active Region Safe Operating Area

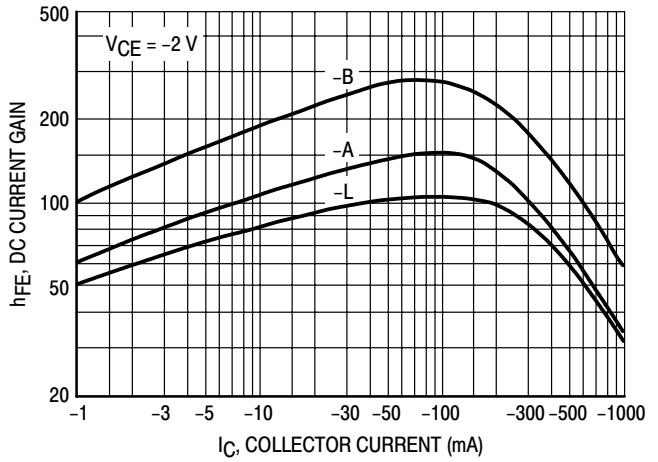


Figure 2. DC Current Gain

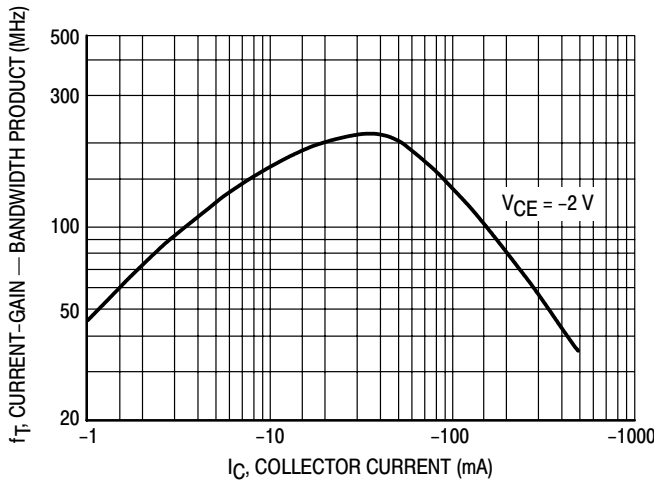


Figure 3. Current Gain Bandwidth Product

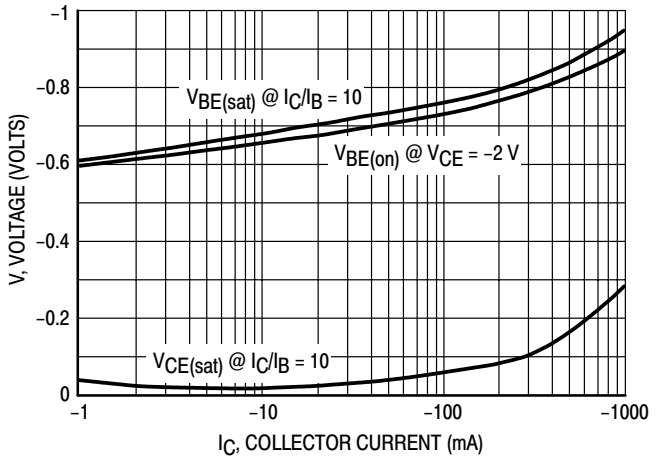


Figure 4. "Saturation" and "On" Voltages

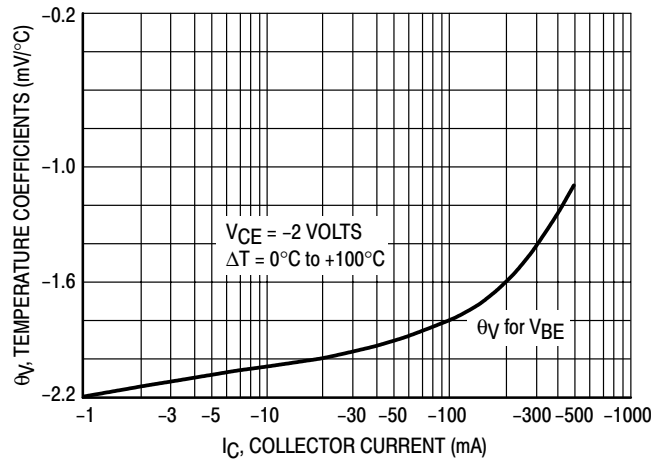
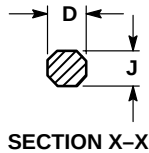
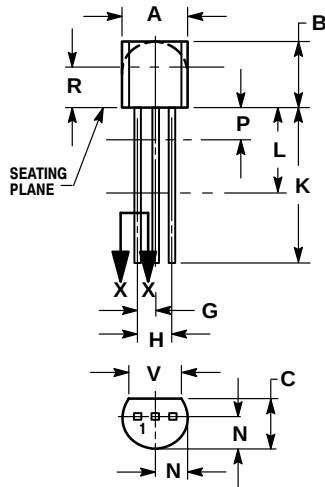


Figure 5. Temperature Coefficients

BC636, BC636-16, BC638, BC640, BC640-16

PACKAGE DIMENSIONS

TO-92
(TO-226)
CASE 29-11
ISSUE AL



NOTES:

1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
2. CONTROLLING DIMENSION: INCH.
3. CONTOUR OF PACKAGE BEYOND DIMENSION R IS UNCONTROLLED.
4. LEAD DIMENSION IS UNCONTROLLED IN P AND BEYOND DIMENSION K MINIMUM.

DIM	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A	0.175	0.205	4.45	5.20
B	0.170	0.210	4.32	5.33
C	0.125	0.165	3.18	4.19
D	0.016	0.021	0.407	0.533
E	0.045	0.055	1.15	1.39
F	0.095	0.105	2.42	2.66
G	0.015	0.020	0.39	0.50
H	0.500	---	12.70	---
I	0.250	---	6.35	---
J	0.080	0.105	2.04	2.66
K	---	0.100	---	2.54
L	0.115	---	2.93	---
M	0.135	---	3.43	---

STYLE 14:

1. EMITTER
2. COLLECTOR
3. BASE

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